

Computer Systems & Network Architecture

Course: CS-401 Distributed Systems | **Module:** 01 Transport Protocols

1. Introduction to Network Layering

Modern distributed networks rely on hierarchical abstraction models to isolate hardware implementation details from application logic. In packet-switched communication, end hosts decompose data streams into discrete segments, encapsulated across successive network layers before transmission across physical media.

2. Mathematical Formulation of Throughput

The capacity of a data path is governed by the Bandwidth-Delay Product (BDP). The BDP determines the volume of data in flight required to fully saturate transmission links:

$$\text{BDP} = \text{Link Bandwidth (bps)} \times \text{Round Trip Time (RTT in seconds)}$$

Furthermore, for Transmission Control Protocol (TCP) connections encountering packet loss probability p , throughput estimation often follows the Mathis formula: $\text{Throughput} \leq (\text{MSS} / \text{RTT}) \times (C / \sqrt{p})$.

3. Summary of Protocol Stack Layers

Layer Number	Layer Name	Primary Protocol	Data Unit (PDU)
7	Application	HTTP/3, DNS, gRPC	Message
4	Transport	TCP, UDP, QUIC	Segment / Datagram
3	Network	IPv4, IPv6, BGP	Packet
2	Data Link	Ethernet, IEEE 802.11	Frame
1	Physical	Optical Fiber, Copper	Bits

4. Congestion Control & Avoidance Algorithms

Congestion avoidance algorithms regulate transmission rate dynamically in response to buffer overflows and packet drops. Classic algorithms such as TCP Tahoe, Reno, and CUBIC use Additive-Increase / Multiplicative-Decrease (AIMD) mechanisms.

Modern transport frameworks like BBR (Bottleneck Bandwidth and RTT) decouple congestion detection from packet loss, sampling real-time delivery rates to operate near the Kleinrock optimum operating point.

5. Review Questions for Cognitive Evaluation

- **Q1:** Contrast packet loss recovery between TCP selective acknowledgments (SACK) and standard cumulative ACK.
- **Q2:** How does a high Bandwidth-Delay Product affect TCP receiver window buffer sizing?
- **Q3:** Explain why UDP is preferred over TCP for low-latency streaming applications.